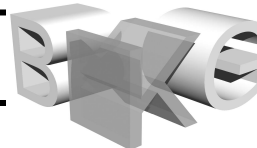


# Green Building Design II – Winter Semester 2026

## Energy Performance Analysis : Berliner Allee "LAB" Student Residence



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### 1. Brief About the building

The analysed building is the student residence Berliner Allee in Darmstadt, Germany, operated by Studierendenwerk Darmstadt. It is a multi-storey student housing building constructed with reinforced concrete and masonry façade elements. Buildings of this type typically exhibit moderate thermal performance compared to current energy efficiency standards.

### 3. Green solutions

| No.                                                                                            | Material layers                         | Thickness d (m) | Thermal conductivity $\lambda$ (W/mK) | Thermal resistance $R = d/\lambda$ (m <sup>2</sup> /K) |
|------------------------------------------------------------------------------------------------|-----------------------------------------|-----------------|---------------------------------------|--------------------------------------------------------|
| Rse                                                                                            | External surface thermal resistance [1] | —               | —                                     | 0.04                                                   |
| 1                                                                                              | Exterior finish coat                    | 0.01            | 0.7 [2]                               | 0.014                                                  |
| 2                                                                                              | Cement render                           | 0.02            | 0.9 [3]                               | 0.022                                                  |
| 3                                                                                              | Mineral wool insulation boards          | 0.08            | 0.04 [4]                              | 2                                                      |
| 4                                                                                              | Load-bearing brick/concrete wall        | 0.175           | 0.77 [5]                              | 0.227                                                  |
| 5                                                                                              | Gypsum board (internal finish)          | 0.015           | 0.25 [6]                              | 0.06                                                   |
| Rsi                                                                                            | Internal surface thermal resistance [1] | —               | —                                     | 0.13                                                   |
| Total of heat transfer resistance ( $R_{tr} = R_{se} + R_1 + R_2 + R_3 + R_4 + R_5 + R_{si}$ ) |                                         |                 |                                       | 2.493m <sup>2</sup> /K                                 |
| Thermal transmittance (U-value = $1/R_{tr}$ )                                                  |                                         |                 |                                       | 0.401=0.40W/m <sup>2</sup> K                           |

Table 2: U-value calculation for external wall

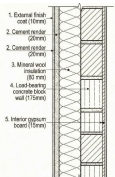


Figure 2: Cross section of the exterior wall (Source: Drawn by Author)

**Compared to GEG**

According to the German Building Energy Act (GEG), the maximum permissible U-value for external walls is 0.24 W/m<sup>2</sup>K. The calculated U-value of 0.40 W/m<sup>2</sup>K exceeds this requirement, indicating that the existing wall construction has insufficient thermal insulation and requires improvement.

| No.                                                                                            | Material layers                                  | Thickness d (m) | Thermal conductivity $\lambda$ (W/mK) | Thermal resistance $R = d/\lambda$ (m <sup>2</sup> /K) |
|------------------------------------------------------------------------------------------------|--------------------------------------------------|-----------------|---------------------------------------|--------------------------------------------------------|
| Rse                                                                                            | External surface thermal resistance [1]          | —               | —                                     | 0.04                                                   |
| 1                                                                                              | Solid wood flooring                              | 0.01            | 0.147 [13]                            | 0.668                                                  |
| 2                                                                                              | Cement screed (density ~1800 kg/m <sup>3</sup> ) | 0.02            | 0.9 [3]                               | 0.022                                                  |
| 3                                                                                              | Mineral wool insulation boards                   | 0.075           | 0.038 [4]                             | 1.974                                                  |
| 4                                                                                              | Reinforced concrete slab                         | 0.13            | 2.1 [5]                               | 0.062                                                  |
| 5                                                                                              | Gypsum board ceiling                             | 0.015           | 0.25 [6]                              | 0.06                                                   |
| Rsi                                                                                            | Internal surface thermal resistance [1]          | —               | —                                     | 0.17                                                   |
| Total of heat transfer resistance ( $R_{tr} = R_{se} + R_1 + R_2 + R_3 + R_4 + R_5 + R_{si}$ ) |                                                  |                 |                                       | 2.396m <sup>2</sup> /K                                 |
| Thermal transmittance (U-value = $1/R_{tr}$ )                                                  |                                                  |                 |                                       | 0.401=0.40W/m <sup>2</sup> K                           |

Table 2: U-value calculation for floor

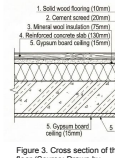


Figure 3: Cross section of the floor (Source: Drawn by Author)

**Compared to GEG**

According to the German Building Energy Act (Gebäudeenergiegesetz - GEG), the maximum permissible U-value for a floor above an unheated room is 0.30 W/m<sup>2</sup>K. The calculated U-value of 0.42 W/m<sup>2</sup>K exceeds this requirement, indicating that additional insulation is required to meet the national energy efficiency standard.

| No.                                                                                                        | Material layers                         | Thickness d (m) | Thermal conductivity $\lambda$ (W/mK) | Thermal resistance $R = d/\lambda$ (m <sup>2</sup> /K) |
|------------------------------------------------------------------------------------------------------------|-----------------------------------------|-----------------|---------------------------------------|--------------------------------------------------------|
| Rse                                                                                                        | External surface thermal resistance [1] | —               | —                                     | 0.04                                                   |
| 1                                                                                                          | Glass pane                              | 0.004           | 1 [10]                                | 0.004                                                  |
| 2                                                                                                          | Argonair gas layer                      | 0.016           | 0.026 [12]                            | 0.615                                                  |
| 3                                                                                                          | Glass pane                              | 0.004           | 1 [10]                                | 0.004                                                  |
| Rsi                                                                                                        | Internal surface thermal resistance [1] | —               | —                                     | 0.13                                                   |
| Total of heat transfer resistance ( $R_{tr} = R_{se} + R_1 + R_2 + R_3 + R_4 + R_5 + R_{si}$ )             |                                         |                 |                                       | 0.739m <sup>2</sup> /K                                 |
| Thermal transmittance (U-value = $1/R_{tr}$ )                                                              |                                         |                 |                                       | 1.353W/m <sup>2</sup> K                                |
| Thermal transmittance of frame: $U_{f,trans} = 1.30$ W/m <sup>2</sup> K (PVC-insulated frame typical) [19] |                                         |                 |                                       |                                                        |
| Linear thermal bridge: $\psi_g = 0.08$ W/mK (typical spacer value) [20]                                    |                                         |                 |                                       |                                                        |
| Glazed area dimensions: 0.54 m × 0.40 m — $A_g = 0.216$ m <sup>2</sup>                                     |                                         |                 |                                       |                                                        |
| Frame area: $A_f = 0.20$ m <sup>2</sup> (simple assumed frame area)                                        |                                         |                 |                                       |                                                        |
| Glass edge length: $l_g = 20.54 - (0.40 + 1.88)$                                                           |                                         |                 |                                       |                                                        |
| Total U-value = $(U_g \times A_g + U_f \times A_f + \psi_g \times l_g) / (A_g + A_f)$                      |                                         |                 |                                       | 1.64W/m <sup>2</sup> K                                 |

Table 3: U-value calculation for door

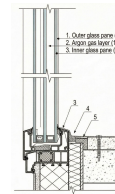
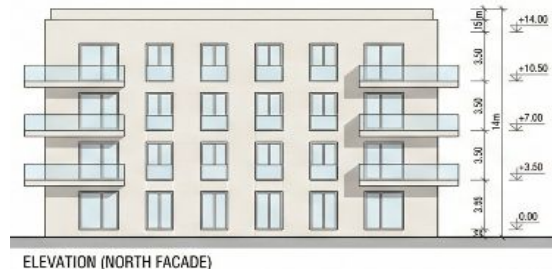
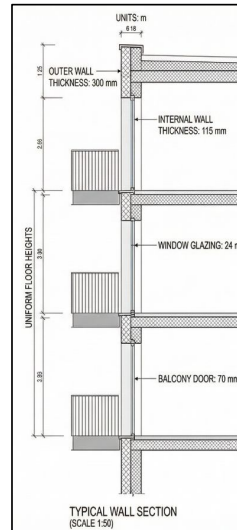
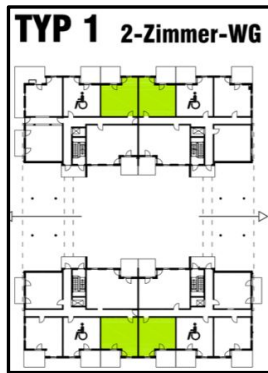


Figure 4: Cross section of the door (Source: HUECK Provider)

**Compared to GEG**

According to the German Building Energy Act (Gebäudeenergiegesetz - GEG), the maximum permissible U-value for new external doors is 1.8 W/m<sup>2</sup>K. The calculated door U-value of 1.64 W/m<sup>2</sup>K meets this requirement, demonstrating compliance with national energy efficiency standards.

### 3. Scaled Plan and Elevations



ELEVATION (NORTH FACADE)

Figure 1: Scaled plan and elevation (Studierendenwerk Darmstadt)

| Element          | Estimated measured thickness |
|------------------|------------------------------|
| External Wall    | 0.3 m                        |
| Internal Wall    | 0.115 m                      |
| Room Door        | 0.04 m                       |
| Window (typical) | ~24 mm                       |
| Balcony Door     | ~70 mm incl. glazing         |

Table 1: Element thicknesses (Self measured)

### 4. Energy Efficiency Calculations

#### 4.2 SOLAR GAINS THROUGH WINDOWS

Germany is located in the Northern Hemisphere where solar altitude varies significantly between summer and winter. During summer, the high sun angle reduces solar penetration through south-facing windows, while in winter the lower solar altitude allows increased solar radiation to enter the building. Therefore, the south-oriented façade was selected to evaluate solar gains through windows using hourly solar radiation data obtained from the PVGIS database.

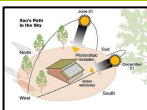


Figure 6: Movement of the sun between the southern and northern hemispheres and its daily east-to-west path (Source: Weerayannu et al., 2017)

| The summer solstice (21/06/2023) – the hours of peak solar radiation intensity |         |         |         |         |         |         |         |        |        |              |
|--------------------------------------------------------------------------------|---------|---------|---------|---------|---------|---------|---------|--------|--------|--------------|
| Universal Time                                                                 | 8:30    | 9:30    | 10:30   | 11:30   | 12:30   | 13:30   | 14:30   | 15:30  | 16:30  | Source       |
| Total Solar radiation intensity $G = G_b + G_d + G_r$ (W/m <sup>2</sup> )      | 253.68  | 327.71  | 329.59  | 363.03  | 399.68  | 396.27  | 276.96  | 146.3  | 48.47  | PVGIS system |
| Window area ( $A_{window}$ ) m <sup>2</sup>                                    | 8.05    |         |         |         |         |         |         |        |        |              |
| Total g-value (1), (2) $g_{total} = g \times F_c$                              | 0.5325  |         |         |         |         |         |         |        |        |              |
| Solar Gain (W)                                                                 | 1087.43 | 1404.77 | 1412.83 | 1556.17 | 1713.28 | 1698.66 | 1187.22 | 627.13 | 207.77 |              |
| Solar = $A_{window} \times g_{total} \times G$                                 | 1087.43 | 1404.77 | 1412.83 | 1556.17 | 1713.28 | 1698.66 | 1187.22 | 627.13 | 207.77 |              |
| The winter solstice (21/12/2023) – the hours of peak solar radiation intensity |         |         |         |         |         |         |         |        |        |              |
| Universal Time                                                                 | 8:30    | 9:30    | 10:30   | 11:30   | 12:30   | 13:30   | 14:30   | 15:30  | 16:30  | Source       |
| Total Solar radiation intensity $G = G_b + G_d + G_r$ (W/m <sup>2</sup> )      | 2.69    | 9.8     | 11.25   | 9.16    | 6.49    | 5.33    | 3.47    | 0      | 0      | PVGIS system |
| Window area ( $A_{window}$ ) m <sup>2</sup>                                    | 8.05    |         |         |         |         |         |         |        |        |              |
| Total g-value (1), (2) $g_{total} = g \times F_c$                              | 0.5325  |         |         |         |         |         |         |        |        |              |
| Solar Gain (W)                                                                 | 11.53   | 42.01   | 48.22   | 39.27   | 27.82   | 22.85   | 14.87   | 0      | 0      |              |
| Solar = $A_{window} \times g_{total} \times G$                                 | 11.53   | 42.01   | 48.22   | 39.27   | 27.82   | 22.85   | 14.87   | 0      | 0      |              |

Figure 6: Solar gains through windows on the Summer Solstice (21/06/2023) and Winter Solstice (21/12/2023) (Source: Author).

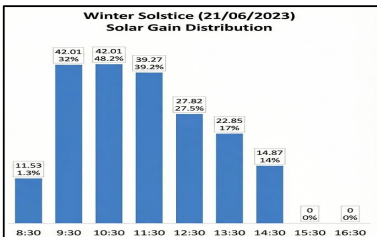
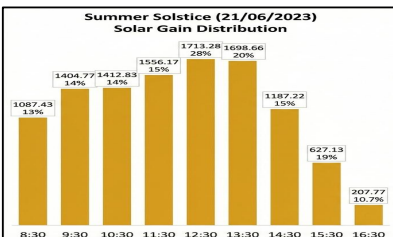


Figure 6: Solar gains through windows on the Summer and winter Solstice (Source: Author)

Solar radiation on building façades varies throughout the year due to seasonal changes in solar altitude in the Northern Hemisphere. In summer, the higher sun angle reduces direct solar penetration through south-facing windows, whereas in winter the lower solar altitude allows deeper solar radiation entry. Therefore, the south-oriented façade was analysed to evaluate solar heat gains using hourly radiation data from the Photovoltaic Geographical Information System (PVGIS) for Darmstadt, Germany [1]. Results indicate that solar gains peak around midday during the summer solstice due to higher irradiance levels, while winter gains are lower overall but occur when direct radiation reaches the vertical façade. Total solar radiation GGG represents the sum of direct beam radiation ( $G_b$ ), diffuse sky radiation ( $G_d$ ), and ground-reflected radiation ( $G_r$ ), automatically calculated by PVGIS for the selected surface orientation. (4) A constant glazing g-value of 0.5325 was assumed based on typical double-glazed window performance for residential buildings.

(1) Solar radiation data were obtained from the Photovoltaic Geographical Information System (PVGIS) using the ERA5 climate database for Darmstadt, Germany. (2) The analysed façade was modelled as a vertical south-oriented surface with a slope of 90° and an azimuth angle of 0°, corresponding to the main window orientation of the building. (3) Total solar radiation intensity GGG represents the sum of direct beam radiation ( $G_b$ ), diffuse sky radiation ( $G_d$ ), and ground-reflected radiation ( $G_r$ ), automatically calculated by PVGIS for the selected surface orientation. (4) A constant glazing g-value of 0.5325 was assumed based on typical double-glazed window performance for residential buildings.

#### 4.2 SOLAR GAINS THROUGH WINDOWS 4.3 SOLAR PANEL ENERGY PRODUCTION ESTIMATE

Estimation of Solar Energy Production for a Residential Building in Haiger, Germany Using PVGIS

##### Input Parameters for PVGIS

| Indicator                                                                                                               | Value                     |
|-------------------------------------------------------------------------------------------------------------------------|---------------------------|
| (Roof footprint area (m <sup>2</sup> ) 2,288.04 (Google Maps measurement) [22])                                         |                           |
| Roof area available for installation Estimated usable roof area assuming 70–80% roof usability for PV installation [20] | ≈ 1,716 m <sup>2</sup>    |
| Area of one solar panel                                                                                                 | 1.84 m <sup>2</sup> [21]  |
| Number of panels                                                                                                        | ≈ 932 (= 1,716.03 / 1.84) |
| Power capacity of one panel                                                                                             | 0.4 kWp [21]              |
| Total installed capacity                                                                                                | ≈ 373 (= 932 × 0.40)      |

Table 6: General data

| Parameter               | Value                |
|-------------------------|----------------------|
| Location                | Darmstadt            |
| Database                | PVGIS-SARAH3         |
| PV technology           | Crystalline silicon  |
| Installed peak PV power | 373 kWp              |
| System losses:          | (PVGIS default) [15] |
| Slope (tilt)            | 35° [15]             |
| Azimuth:                | 0° (South) [15]      |

Table 7: General data

##### Result from PVGIS

|                                                   |           |
|---------------------------------------------------|-----------|
| Simulation outputs:                               |           |
| Slope angle [°]                                   | 35        |
| Azimuth angle [°]                                 | 0         |
| Yearly PV energy production (kWh)                 | 42,006.65 |
| Yearly in-plane irradiation (kWh/m <sup>2</sup> ) | 1405.95   |
| Year-to-year variability (kWh)                    | 18510.46  |
| Changes in output due to:                         |           |
| Angle of incidence [%]                            | -2.83     |
| Spectral effects [%]                              | 1.73      |
| Temperature and low irradiance [%]                | -5.32     |
| Total loss [%]                                    | -19.52    |

Figure 7: Simulation outputs (Source: Author's calculation using PVGIS)

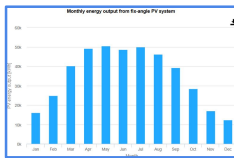


Figure 8: Monthly energy output from fixed PV system (Source: Author's calculation using PVGIS)

The PVGIS simulation indicates that monthly electricity production is highest between April and August, reflecting higher solar irradiation and longer daylight periods in summer. Winter months show reduced output due to lower sun angles and shorter days. Overall, the results highlight a strong seasonal pattern in PV electricity generation (PVGIS, European Commission JRC) [15].

##### Estimated Solar Coverage per Residential Unit

###### Result from PVGIS

400,000 kWh per year [16]  
Building electricity demand  
(~250 resident units)

422,006.85 kWh/year  
Annual PV electricity production  
(~470 solar panels)

Percentage of solar electricity production relative to building demand

$(422,006.85 \text{ kWh}) \div (400,000 \text{ kWh}) \times 100\% = 105.5\%$

The installed rooftop photovoltaic system can supply approximately 105.5% of the annual electricity demand of the Studierendenwerk Berlin from Berliner Allee residence. The remaining electricity demand must be covered by grid supply, while solar generation significantly reduces operational carbon emissions and energy costs.

The photovoltaic installation demonstrates significant potential for renewable energy integration in student housing buildings. With an estimated annual production of 262,232 kWh, the rooftop PV system can cover approximately 66% of the building's electricity demand. This contribution substantially improves energy performance and supports climate mitigation targets by reducing dependence on conventional grid electricity.

### 5. Reference

- [1] DIN EN ISO 6946. Building components and building elements – Thermal resistance and thermal transmittance.
- [2] European plaster/render material thermal property database.
- [3] Cement render plaster material thermal property database.
- [4] ISO 10456. Thermal properties of building materials and products.
- [5] Concrete and masonry thermal conductivity reference values.
- [6] Gypsum board material thermal performance data sheets.
- [7] EN 673. Glass in building – Determination of thermal transmittance.
- [8] Argon gas thermal conductivity data. ISO glazing standards.
- [9] Passive House Institute. Thermal bridge coefficients for window frames.
- [10] German Building Energy Act (GEG). Maximum permissible U-values.
- [11] Studierendenwerk Darmstadt. Student Residence Berliner Allee building information.
- [12] Beck et al. Solar geometry and seasonal sun path diagram. ResearchGate figure: [https://www.researchgate.net/figure/The-movement-of-the-sun-from-the-south-to-northern-hemisphere-and-along-the-latitude-form\\_fig5\\_315381369](https://www.researchgate.net/figure/The-movement-of-the-sun-from-the-south-to-northern-hemisphere-and-along-the-latitude-form_fig5_315381369)
- [13] European Commission Joint Research Centre (JRC). PVGIS Solar Radiation Database.
- [14] PVGIS User Manual and calculation methodology.
- [15] European Commission JRC. PVGIS Photovoltaic Geographical Information System: <https://re.jrc.ec.europa.eu/pvgis/>
- [16] Google Maps. Distance and area measurement tool, Berliner Allee Student Residence, Darmstadt.
- [17] Typical crystalline silicon photovoltaic panel technical specifications (~400 W module).
- [18] German Federal Ministry for Economic Affairs and Climate Action (BMWK). Household electricity consumption statistics.
- [19] European Commission – PVGIS Photovoltaic Geographical Information System. [https://re.jrc.ec.europa.eu/pvgis\\_tools/](https://re.jrc.ec.europa.eu/pvgis_tools/)